

AI-POWERED GEOSPATIAL RECONSTRUCTION PLATFORM

LISS-IV COMPOSITE INPAINTING & DYNAMIC CORES

ANALYSIS REPORT

ACQUISITION SESSION ID : test-session-uuid-11b
LOCK TARGET DATASET : R2F14JUL2025073870009500051SSANSTUC00GTDB
GENERATION TIMESTAMP : 2026-06-28 21:46:45 UTC
SECURITY REGISTRY CODE : ISRO-BAH26-PROD-MC
SYSTEM READINESS LEVEL : OPERATIONAL STATUS OK

CONFIDENTIALITY NOTICE: This report contains operational earth telemetry details.
Prepared exclusively for scientific research, analysis, and operations validation.

// EXECUTIVE SUMMARY & CORE DIAGNOSTICS

DATASET STATUS: COMPLETED REGISTRATION DATE: 2026-06-28 16:16:45 UTC
SENSOR MODULE: LISS-IV Multi-spectral TARGET SESSION ID: test-session-uuid-11b

Ingested telemetry node verified. Baseline spatial inpainting completed successfully.
Post-optimization spectral blending active. Average reliability index calculated at 92.5%.

// LISS-IV DATASET INGESTION & METADATA DETAILS

COORDINATE CRS:	UTM	PROJECTION MODEL:	Transverse Mercator
SPATIAL RES (X):	5.00m	SPATIAL RES (Y):	5.00m
RASTER WIDTH:	2048 px	RASTER HEIGHT:	2048 px
BAND INGEST COUNT:	3	METADATA INTEGRITY:	LISS-IV SIM LOCKED SECURE

Extracted structural spatial metrics from satellite headers. The geometric footprint matches LISS-IV sensor resolutions, locking the CRS system.

// GEOSPATIAL SPATIAL CONTEXT & CRS PROFILE

CENTER LATITUDE:	12.000000	CENTER LONGITUDE:	80.000000
LATITUDE BOUNDS:	11.90000 to 12.10000	LONGITUDE BOUNDS:	79.90000 to 80.10000
REFERENCE CRS:	EPSG:32644		

Calculated absolute latitudinal and longitudinal boundaries of the footprint to calibrate candidate satellite discovery.

// TEMPORAL DISCOVERY STACK REFERENCES

REFERENCES DISCOVERY:	3 candidates resolved	MEAN CLOUD COVER:	15.00%
MEAN OVERLAP SCORE:	90.00%	MEAN TEMP DISTANCE:	20.0 days

Temporal context loaded from 3 low-cloud reference scenes.

// CLOUD SEGMENTATION & TARGET RECONSTRUCTION MASKS

SEGMENTED REGIONS:	10	CLOUD AREA PIXEL COUNT	5000 px
SHADOW AREA PIXEL COUNT	2000 px	OBSCURED PERCENTAGE:	12.50%

Completed class-based cloud segmentation. Binary mask targets have been successfully parsed for generative inpainting boundaries.

// GENERATIVE AI RECONSTRUCTION CONFIGURATIONS

INPAINTING STATUS:	COMPLETED	PRIMARY METHOD:	cv2.INPAINT_TELEA
OPTIMIZATION STATE:	COMPLETED	EXECUTION DURATION:	1200 ms

Baseline reconstruction compiled via temporal-weighted inpainting.

// AI RECONSTRUCTION QUALITY SCORECARD

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OVERALL QUALITY SCORE: 94.2 / 100

SPECTRAL SIMILARITY INDEX: 93.8%
SPATIAL BOUNDARY COHERENCE: 95.1%
INPAINTED MASK COVERAGE: 100.0%
OPTIMIZATION BLENDING: Active

// STATISTICAL RELIABILITY & CONFIDENCE SCORES

ESTIMATION STATUS: completed AVERAGE CONFIDENCE INDEX: 92.50%
LOW RELIABILITY COVERAGE: 1.20% SCORING STRATEGY: Temporal/Spectral Agreement Index
Calculated standard deviations across candidates.

[INFO] Reliability Calibration Alert: Overall Reliability Score: 92.5% (Tier: High).